

Steven Kolawole

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RESEARCH FOCUS

The efficiency of LLM inference: model compression, KV cache optimization, adaptive serving (cascading and speculative decoding), and inference parallelism for production systems.

EDUCATION

Ph.D., Computer Science (Language Technologies), Carnegie Mellon University, USA 2023 – 2028 (expected)

Language Technologies Institute, School of Computer Science. Advisor: Prof. Virginia Smith.

B.Sc., Computer Science, Federal University of Agriculture Abeokuta, Nigeria 2017 – 2023

RESEARCH EXPERIENCE

Applied Scientist Intern, AWS AGI Foundations May – Aug 2026

San Francisco, CA. Inference Optimization Science Team (Bedrock); mentors: Drs. Nathan Pemberton & Kyle Ulrich. In-place speculative decoding via gated LoRA adapters, mask priming, and verifier-gated loss in quadratic and draft-then-verify topologies.

Graduate Student Researcher, Carnegie Mellon University Aug 2023 – Present

Pittsburgh, PA. Advised by Prof. Virginia Smith. Algorithmic methods for efficient ML inference.

Independent Researcher, ML Collective Apr 2021 – Present

Remote. Pre-Ph.D. research mentored by Drs. Rosanne Liu & Jason Yosinski; now mentoring junior researchers within the same open-science pipeline. Resource-constrained NLP, federated learning, and efficient ML.

PUBLICATIONS

* denotes equal contribution.

- **S. Kolawole**, K. Santhanam, V. Smith, P. Thaker. *ParallelPrompt: Extracting Parallelism from Large Language Model Queries*. NeurIPS 2025, Datasets & Benchmarks Track. [OpenReview](#) · [arXiv:2506.18728](#).
- D. Soiffer, **S. Kolawole**, V. Smith. *Semantic Agreement Enables Efficient Open-Ended LLM Cascades*. EMNLP 2025, Industry Track. [ACL Anthology](#) · [arXiv:2509.21837](#).
- **S. Kolawole***, D. Dennis*, A. Talwalkar, V. Smith. *Agreement-Based Cascading for Efficient Inference*. Transactions on Machine Learning Research (TMLR), 2025. [OpenReview](#) · [arXiv:2407.02348](#).
- **S. Kolawole***, L. Dery*, J.-F. Kagy, V. Smith, G. Neubig, A. Talwalkar. *Everybody Prune Now: Structured Pruning of LLMs with only Forward Passes (Bonsai)*. [arXiv:2402.05406](#).
- **S. Kolawole**, O. Osakuade, N. Saxena, B. K. Olorisade. *Sign-to-Speech Model for Sign Language Understanding: A Case Study of Nigerian Sign Language*. IJCAI 2022, AI for Social Good Track. Proceedings · [arXiv:2111.00995](#).

MANUSCRIPTS UNDER REVIEW

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- **S. Kolawole**, V. Smith. *Epiphany-Aware KV Cache Eviction Without the Attention Matrix (EpiKV)*. Under review, ICLR 2027. [arXiv:2606.26472](#).
 - P. Jim*, **S. Kolawole***, O. M. Busoye, G. Bagai, V. Smith. *What Survives When You Compress a Recursive Reasoner for the Edge?* (EdgeTRM). Under review, ICLR 2027. [arXiv:2606.26488](#).

WORKSHOP PAPERS

- M. Oduwale, . . . , A. Owodunni, **S. Kolawole**. *From Scarcity to Efficiency: Investigating the Effects of Data Augmentation on African Machine Translation*. LM4UC @ AAAI 2026 (Springer CCIS). [OpenReview](#). **[senior/last author]**
- N. Obiefuna, S. Oyeneye*, S. Odunaiya*, I. Oyelaja*, **S. Kolawole**. *PrivacyBench: Privacy Isn't Free in Hybrid Privacy-Preserving Vision Systems*. ES-FoMo @ ICML 2025. [OpenReview](#). **[senior/last author]**
- B. Awobade*, M. Oduwale*, **S. Kolawole**. *What Happens When Small Is Made Smaller? Exploring the Impact of Compression on Small Data Pretrained Language Models*. AfricaNLP @ ICLR 2024. [arXiv:2404.04759](#). **[senior/last author]**
- C. Leong, . . . , **S. Kolawole**, et al. *Adapting to the Low-Resource Double-Bind: Investigating Low-Compute Methods on Low-Resource African Languages*. AfricaNLP @ ICLR 2023. [arXiv:2303.16985](#).
- N. Alam*, **S. Kolawole***, S. Sethi*, N. Bansali, K. Nguyen. *Vision Transformers for Mobile Applications: A Short Survey*. [arXiv](#), 2023. [arXiv:2305.19365](#).

PEER REVIEW & PROFESSIONAL SERVICE

- **Machine learning / AI**: NeurIPS (2024, 2026); ICML (2026); ICLR (2025); AISTATS (2025, 2026); AAAI (2026).
- **Language modeling / NLP**: COLM (2025, 2026); ACL System Demonstrations (2025, 2026); EMNLP System Demonstrations (2024, 2025, 2026).
- **Workshop**: Efficient Systems for Foundation Models (ES-FoMo) @ ICML (2024).

HONORS & AWARDS

- **Scholar, Machine Learning Summer School (MLSS) NYC** (Columbia University) — competitively selected among ~200 Ph.D. researchers worldwide; fully funded. 2026
- **Research Grant, Algorand Foundation** — \$114,635 Ecosystem grant for an ML research-engineering project (the ASAnalytics platform). 2022
- **National Artificial Intelligence Champion**, Nigeria Computer Society — first place, national AI Challenge. 2022
- **Ideathon Winner** (\$5,000 Google Cloud credits), Deep Learning Indaba — the continent's foremost ML gathering. 2022
- **Best Poster, Data Science Nigeria**. 2021
- **Winner, DeepQuest AI Challenge**. 2021
- **Scholar, MTN Foundation Science & Technology Scholarship** — selection rate 1.8%. 2020, 2021
- **Led AI+ Abeokuta to Data Science Nigeria's "AI City of the Year"** — as community lead, produced the largest national-bootcamp cohort of any Nigerian city. 2020
- **Runner-up, "Mr. Algorithm," Data Science Nigeria** — nationwide community vote, for technical excellence and knowledge-sharing. 2020
- **Fellow, Young Professionals Bootcamp** — selection rate 3.6%. 2020
- **Gold Medal, NaijaHacks** — AgroAI plant-disease-detection app; top 12 of 336 teams. 2019

LEADERSHIP & COMMUNITY SERVICE

- **Founder & Lead, ML Collective Africa** 2023 – Present
Conceived, founded, and lead the Africa program of ML Collective (a 501(c)(3) research non-profit). Operate as the program's research lead: set the research agenda, lead weekly research meetings, design structured study programs, advise member projects, and develop a team of organizers and mentors. Administer ML Collective's GCP grant (tens of thousands of dollars) — reviewing project proposals and allocating compute. Raised over \$50,000 since 2022 to fund African researchers' Deep Learning Indaba attendance and the program's operations, and chair the program's accepted workshop at the Deep Learning Indaba 2026 ("Whose Intelligence? Building Agentic AI in Africa without Frontier Model Dependency").
- **Campus Lead, AI+ FUNAAB; lead, AI+ Abeokuta** (Data Science Nigeria) 2019 – 2021
Built a structured remote preparation program that took the community from five national qualifiers (2019) to 38 of 42 entrants selected (2020; 28 in the national top 250, nine in the national top 20).
- **Community Lead, Google Developer Student Clubs, FUNAAB** 2021 – 2022
Technical training reaching 3,000+ students; 600+ taught personally.
- **Mentor, STEM for Development** 2023 – 2025
Mentor 20+ graduate-school applicants per year toward U.S. and other leading research programs.
- **Mentor, She Code Africa** 2021 – 2022
Mentored women entering machine learning and data science.

SELECTED TEACHING

Guest Lecturer (Model Compression) & Teaching Assistant, 10-718 Spring 2026
Machine Learning in Practice, CMU

Invited Lecturer, AI Saturdays Lagos (AI6) Fall 2023

Machine-learning lecture, re-used in every cohort since (available on YouTube); authored the cohort's Python Refresher course.

SELECTED INVITED TALKS & PRESENTATIONS

(Research orals marked †; invited talks/panels noted.)

- **2026:** "Doing ML Research from the Margins," skills session, Deep Learning Indaba (Lagos); "A SOTA Overview of Efficient AI, and What It Means for Africa," IndabaX Nigeria (*invited*).
- **2025:** Black in AI ELAI graduate-school talk (*invited*); CMU Africa Research Club (*invited*, ×2).
- **2024:** Low-resource LLM practicals, Deep Learning Indaba (Senegal).
- **2023:** Cohere For AI (C4AI) Independent-Research panel (*invited*); OSCAFEST.
- **2022:** IJCAI[†]; Deep Learning Indaba ML-efficiency talk (Tunisia); Lanfrica Talks (*invited*); EuroPython (Dublin); PyConDE & PyData Berlin; Berlin Buzzwords; Build Stuff; SciPy (Austin, Texas); NCS AI Summit; DataFest Africa; GDG Abeokuta DevFest (*invited*).
- **2021:** NeurIPS ML4D[†]; Masakhane Speaker Series (*invited*); PyData Global; EuroPython; PyCon APAC; Conf42 Python; ML Collective Research Jam (×3).
- **2020:** PyData Global; PyCon Africa.

SELECTED INDUSTRY EXPERIENCE

Founding ML Engineer, ASAnalytics 2022

Backend systems for real-time opinion-mining/analytics of blockchain's digital assets — the vehicle for the

Algorand Foundation research-engineering grant.

Data Science Intern, SeqHub Analytics LLC, New Haven, CT 2021

Fine-tuned mT5 for low-resource Nigerian-language machine translation; built microservices and a dockerized analytics dashboard.

Machine Learning Intern, Rural Farmers Hub, Abuja, Nigeria 2020

Annotated farms from satellite imagery and built soil-nutrient prediction models for crop recommendations (87% accuracy).

SELECTED MEDIA

Interview on the sign-to-speech research, [Alhub](#) (2022); talk recordings on [PyVideo](#).